### 🧪 Section: Our Dynamic Universe

#### 🔷 Topic: Motion, Equations and Graphs

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**Question 1 – A Van Braking at a Traffic Light**

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**Given:**

- Initial speed: $u = 13.4\,\mathrm{m/s}$
- Final speed: $v = 0\,\mathrm{m/s}$
- Acceleration: $a = -2.85\,\mathrm{m/s^2}$

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**(a) Calculate the distance travelled by the van during braking**

✔3 marks

Use:
$v^2 = u^2 + 2as$

$0 = (13.4)^2 + 2(-2.85)s$

$$0 = 179.56 - 5.7s \quad \Rightarrow \quad s = \frac{179.56}{5.7} = 31.5\,\mathrm{m}$$

✅ **Answer: 31.5 m**

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**(b) Calculate the time taken for the van to come to rest**

✔2 marks

Use:
$v = u + at \Rightarrow 0 = 13.4 + (-2.85)t$

$$t = \frac{-13.4}{-2.85} = 4.70\,\mathrm{s}$$

✅ **Answer: 4.70 s**

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**(c) Sketch velocity-time graph**

✔2 marks

🖊 On the graph:

- Vertical axis (v): from $v = 13.4$ m/s to 0
- Horizontal axis (t): from $t = 0$ to $t = 4.7$ s
- Straight **sloping line down** from (0, 13.4) to (4.7, 0)

✅ Must include:

- Correct linear slope
- Numerical labels on both axes

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#### 🧠 Revision Tips

- Choose equations with only known quantities
- Negative acceleration = **deceleration**
- A **v–t graph** with a sloped line = **uniform acceleration/deceleration**
- Area under v–t graph = **distance**